

TRACY, MN, USA

WMO: 722342

Lat: 44.249N

Lon: 95.607W

Elev: 408

StdP: 96.51

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-21.0	-27.0	0.3	-22.7	-24.9	0.4	-20.7	14.7	-7.4	13.1	-8.7	4.0	320	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.1	23.0	30.5	22.1	29.0	21.3	25.0	29.9	23.9	28.6	22.8	27.2	5.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	19.2	28.2	22.4	18.0	27.1	21.3	16.7	26.0	78.5	29.9	73.7	28.6	69.4	27.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	11.0	9.6	DB	-26.5	35.1	2.8	1.4	-28.5	36.1	-30.2	36.8	-31.8	37.6	-33.8	38.6	
			WB	-26.8	26.6	2.6	1.4	-28.6	27.6	-30.1	28.4	-31.6	29.3	-33.5	30.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-8.1	-7.1	0.6	7.7	15.0	20.5	22.7	21.0	17.4	10.3	1.7	-5.6
	DBStd	12.35	7.44	6.97	7.09	5.95	5.09	3.53	3.14	2.92	4.63	5.31	6.65	6.52
	HDD10.0	2273	561	479	301	109	14	0	0	2	62	259	485	
	HDD18.3	4131	819	713	551	319	127	18	4	8	74	255	499	743
	CDD10.0	1575	0	0	9	42	171	314	395	340	223	71	10	0
	CDD18.3	390	0	0	1	1	25	83	141	90	44	6	0	0
	CDH23.3	3293	0	0	2	22	269	692	1241	644	368	53	1	0
	CDH26.7	971	0	0	0	3	86	209	402	158	104	9	0	0
(14) Wind	WSAvg	4.3	5.1	4.9	4.7	5.0	4.6	3.6	3.2	2.8	3.7	4.3	5.0	4.7
(15) Precipitation	PrecAvg	731	19	18	39	74	88	113	87	89	88	59	36	23
	PrecMax	1042	49	39	92	201	167	279	164	160	295	154	114	83
	PrecMin	540	2	1	3	9	27	57	14	19	17	8	0	2
	PrecStd	124	14	10	23	40	44	47	37	40	66	42	33	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	13.6	22.3	26.1	32.6	33.2	34.1	32.3	32.1	27.8	21.1	10.8
		MCWB	4.0	10.3	13.8	14.9	18.7	22.5	24.5	24.3	21.6	16.9	12.8	8.3
	2%	DB	5.0	7.7	17.2	22.7	29.0	31.0	32.3	30.3	29.8	24.7	17.0	6.9
		MCWB	2.5	4.8	11.6	13.7	17.8	21.0	24.3	23.6	21.0	15.7	11.3	4.1
	5%	DB	2.9	4.4	13.2	20.0	26.6	29.1	30.7	28.4	27.5	21.6	13.8	3.9
		MCWB	1.4	2.1	9.6	11.9	17.2	20.4	23.5	22.2	19.1	14.4	8.8	2.0
	10%	DB	1.3	2.1	10.2	17.2	23.7	27.4	28.9	27.0	25.2	18.5	10.9	2.3
		MCWB	0.2	0.6	7.6	11.0	15.5	19.6	22.3	20.9	18.5	12.5	7.2	0.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	10.0	15.6	16.7	21.8	25.1	27.2	26.1	24.5	19.7	14.4	8.7
		MCDB	7.2	13.7	19.9	22.3	28.3	30.1	31.4	30.8	29.0	24.7	19.0	12.1
	2%	WB	2.8	5.1	12.3	15.1	19.9	23.2	25.4	24.4	22.8	16.9	11.6	3.9
		MCDB	4.7	7.8	15.6	20.9	25.8	28.8	30.4	28.8	26.7	22.4	16.7	5.9
	5%	WB	1.6	2.3	10.0	13.4	18.4	21.8	24.3	23.2	21.1	15.2	9.3	2.2
		MCDB	3.0	4.2	13.9	18.2	24.1	26.9	29.5	27.3	25.6	20.1	13.3	3.8
	10%	WB	0.3	0.6	7.4	11.7	17.0	20.5	23.2	22.0	19.6	13.2	6.9	1.0
		MCDB	1.2	1.8	10.2	16.1	21.8	25.5	27.8	25.8	23.8	17.7	10.7	2.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	8.5	10.3	10.8	10.1	10.3	9.9	11.3	10.5	8.9	7.6
		MCDBR	9.0	10.9	13.3	15.3	14.5	12.4	11.2	11.0	13.0	13.7	12.7	8.9
		MCWBR	7.4	8.0	8.6	8.3	6.7	5.7	5.6	5.8	6.1	7.3	7.6	7.0
	5% WB	MCDBR	8.6	10.4	12.6	13.2	11.9	11.0	10.5	9.9	11.5	12.1	12.3	8.1
		MCWBR	7.0	7.8	8.4	8.3	6.3	6.3	5.9	5.9	6.5	7.2	7.6	6.5
(40) Clear-Sky Solar Irradiance	taub		0.261	0.273	0.306	0.360	0.379	0.395	0.405	0.406	0.363	0.321	0.283	0.265
	taud		2.459	2.438	2.449	2.335	2.350	2.364	2.320	2.290	2.388	2.495	2.524	2.460
	Ebn at Noon		883	931	936	902	892	875	861	846	860	860	847	838
	Edh at Noon		70	85	97	119	121	120	124	124	103	81	66	63
(44) All-Sky Solar Radiation	RadAvg		1.87	2.92	3.83	4.75	5.48	6.11	6.60	5.44	4.29	2.83	1.88	1.41
	RadStd		0.20	0.32	0.38	0.47	0.40	0.42	0.23	0.43	0.31	0.37	0.15	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page